SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

(192524118)

I R·Akshaya *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

20+26+20+20+8

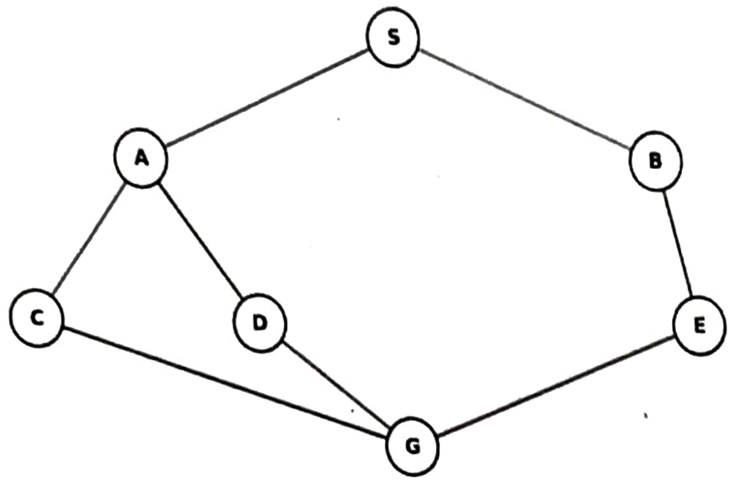
=94

Student 3: Akshaya R | Register Number: 192524118

1. Question 7 - Greedy Best-First Search

Use the displayed graph and assign the following heuristic values: h(A)=6, h(B)=2, h(C)=4, h(D)=1, h(E)=3, h(G)=0.

Perform Greedy Best-First Search from S to G and trace the priority queue.

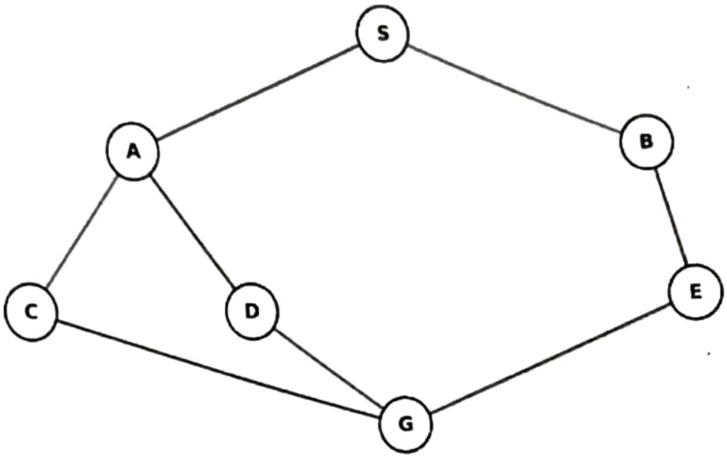


Iteration	Priority Queue	Expanded	h(n)
1			
2			
3			
4			
5			

2. Question 8 - A* Search

Use f(n)=g(n)+h(n). Perform A* search on the displayed graph.

Use the following values: S(0,7), A(2,5), B(3,4), C(4,3), D(5,2), E(6,1), G(8,0).



Iteration	OPEN	Expanded	g	h	f
1					
2					
3					
4					
5					

3. Question 9 - Heuristic Function Analysis

Consider the following table. Determine which heuristic is admissible and explain the consequence of overestimating the true cost.

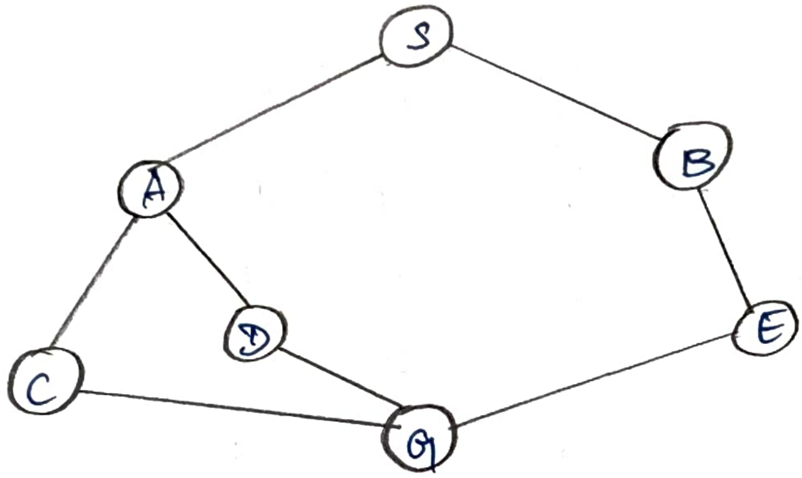
State	Actual Cost	h1(n)	h2(n)
A	10	8	12
B	7	5	6
C	12	10	15
D	5	4	3

1 (7) **GREEDY BEST-FIRST SEARCH**

HEURISTIC VALUES :

$h(A)=6$, $h(B)=2$, $h(C)=4$, $h(D)=1$,
$h(E)=3$, $h(G)=0$.

Perform Greedy Best-First search from S to G and trace priority queue



Greedy Best-First always selects the node with lowest $h(n)$.

Iteration	Priority Queue	Expanded	h(n)
		S	-
1	B(2), A(6)	B	2
2	E(3), A(6)	E	3
3	G(0), A(6)	G	0
4	-		

Path found :

$S \rightarrow B \rightarrow E \rightarrow G$.

Answer : Goal G is reached after Expanding S, B, E, G.

2(8) A^*- Search.

Use $f(n) = g(n) + h(n)$

Given:

Node	g	h	f
S	0	7	7
A	2	5	7
B	3	4	7
C	4	3	7
D	5	2	7
E	6	1	7
G	8	0	8.

Use the graph edge costs implied by the given g-values, the optimal routes have cost 8.

One valid A^* tree, using tie-breaking order B before A, is:

Iteration	OPEN	Expanded	g	h	f
1	A(7), B(7)	S	0	7	7
2	E(7), A(7)	B	3	4	7
3	A(7), G(8)	E	6	1	7
4	C(7), D(7), G(8)	A	2	5	7
5	D(7), G(8)	C	4	3	7

Continuing, G is selected with f = 8.

Optimal cost = 8

Possible optimal paths:

$S \rightarrow B \rightarrow E \rightarrow G$

or

$S \rightarrow A \rightarrow C \rightarrow G$

or

$S \rightarrow A \rightarrow D \rightarrow G$.

Q19) Heuristic Function Analysis:

State	Actual cost	$h_1(n)$	$h_2(n)$
A	10	8	12
B	7	5	6
C	12	10	15
D	5	4	3

$h_1(n)$

$\Rightarrow$ A: $8 \leq 10$

$\Rightarrow$ B: $5 \leq 7$

$\Rightarrow$ C: $10 \leq 12$

$\Rightarrow$ D: $4 \leq 5$

Therefore, h_1 is admissible

$h_2(n)$

$\Rightarrow$ A: $12 > 10$ ✗

$\Rightarrow$ B: $6 \leq 7$

$\Rightarrow$ C: $15 \geq 12$ ✗

$\Rightarrow$ D: $3 \leq 5$

$\therefore$ h_2 is not admissible.

Consequence of Overestimating:

An overestimating heuristic can make A^* ignore the actual optimal path & may produce a non-optimal solution.